Practice workbook · Sequence and describe the stages of mitosis

Ordering prophase, metaphase, anaphase and telophase, describing chromosome behaviour in each, and drawing or recognising metaphase.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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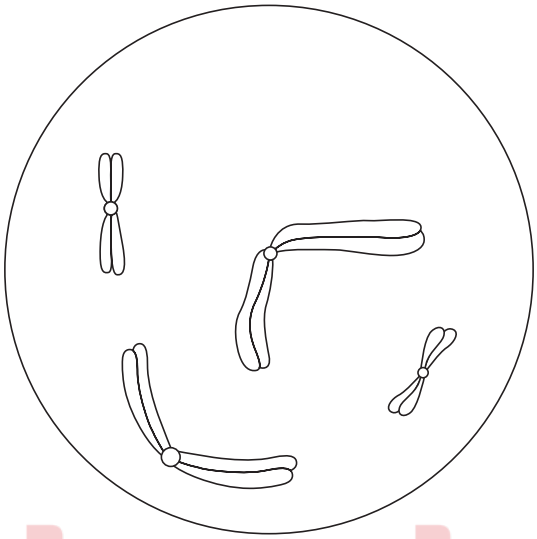
Type ID: HBI-SEQUENCE-MITOSIS-STAGES

1. Question 5 · part b · 5 marks

June 2021 Paper 2

Worked-example retry · Attempt part b only.

5 The diagram shows a cell in the early stages of mitosis.



(a) (i) Which molecule is found in the chromosomes? **(1)**

- ☒ **A** DNA
- ☒ **B** glucose
- ☒ **C** glycogen
- ☒ **D** mRNA

(ii) Draw a circle around two homologous chromosomes on the diagram. **(1)**

(iii) State how the diagram shows that this is not a human cell. **(1)**

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1. Question 5 · part b · 5 marks

June 2021 Paper 2

Worked-example retry · Attempt part b only.

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(b) Describe what happens to the chromosomes during the four stages of mitosis.

(5)

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(Total for Question 5 = 8 marks)

1. Question 5 · part b · 5 marks · Working and answer

Working and answer

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2. Question 6 · part a.iv · 3 marks

November 2020 Paper 1R

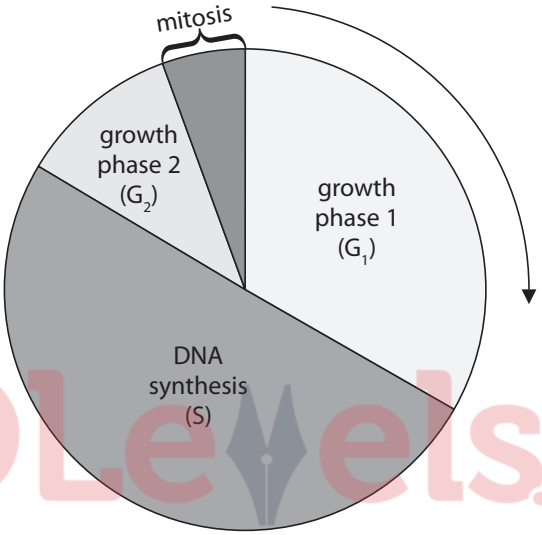
Further practice · Attempt part a.iv only.

6 (a) Cells in the body go through several cell cycles in their lifetime.

Each cell cycle is divided into different stages.

Different events take place in the cell during each stage.

The diagram shows one cell cycle of a body cell.



(i) One cell cycle takes 24 hours to complete.

The total time spent in G_1 and G_2 is ten hours.

Calculate the time spent in mitosis.

(2)

time = ... hours

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2. Question 6 · part a.iv · 3 marks

November 2020 Paper 1R

Further practice · Attempt part a.iv only.

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(ii) Each of the phases of the cell cycle prepare the cell for mitosis.

Suggest the events that take place in the growth phases of the cell cycle.

(2)

(iii) The cell cycle completes by producing two new body cells.

Describe the genetic features of the two cells produced by mitosis.

(2)

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(iv) The table lists some of the stages of mitosis.

Complete the table by giving details of the event that takes place in each stage.

(3)

Stage	Events
metaphase	
anaphase	
telophase	

2. Question 6 · part a.iv · 3 marks

November 2020 Paper 1R

Further practice · Attempt part a.iv only.

(b) Describe the role of DNA polymerase in the synthesis of new DNA molecules.

(2)

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(Total for Question 6 = 11 marks)

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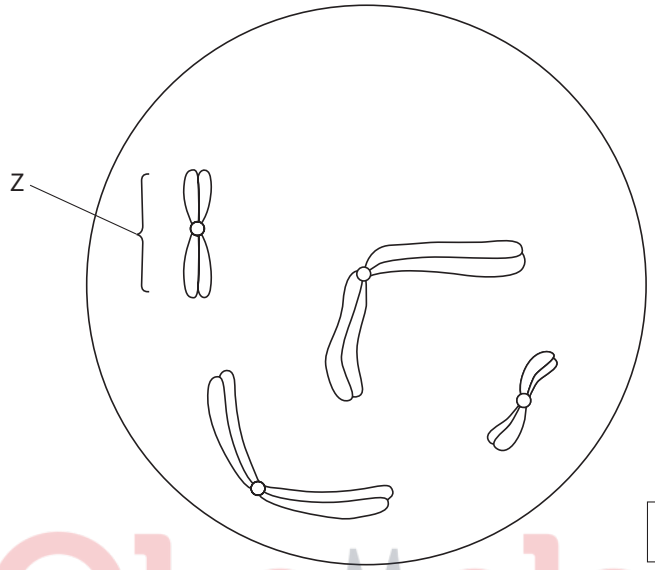
Working and answer

3. Question 7 · part a.i · 2 marks

November 2021 Paper 1

Further practice · Attempt part a.i only.

7 The diagram shows a cell in early stages of mitosis.



(a) The four stages of mitosis are:

- anaphase
- metaphase
- prophase
- telophase

(i) Give these stages in the correct order.

(2)

.....

.....

(ii) Draw a diagram of the cell as it would appear at metaphase.

(3)

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3. Question 7 · part a.i · 2 marks · Working and answer

Working and answer

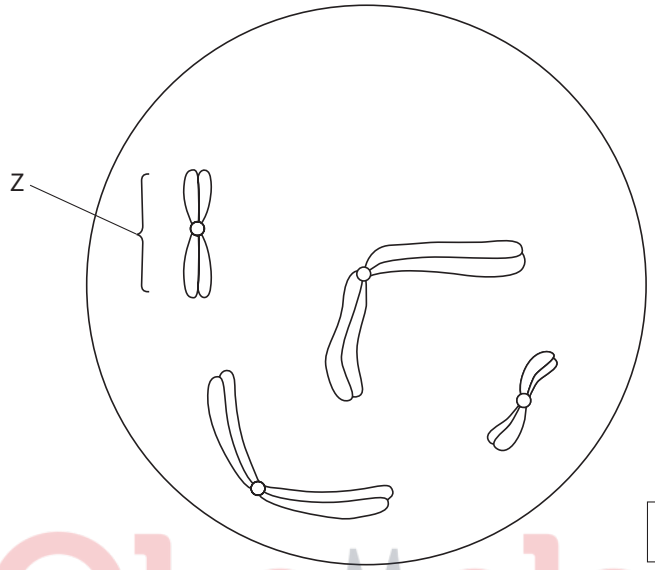
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4. Question 7 · part a.ii · 3 marks

November 2021 Paper 1

Further practice · Attempt part a.ii only.

7 The diagram shows a cell in early stages of mitosis.



(a) The four stages of mitosis are:

- anaphase
- metaphase
- prophase
- telophase

(i) Give these stages in the correct order.

(2)

.....

.....

(ii) Draw a diagram of the cell as it would appear at metaphase.

(3)

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P 6 7 0 6 0 R R A 0 1 6 2 0

4. Question 7 · part a.ii · 3 marks

November 2021 Paper 1

Further practice · Attempt part a.ii only.

(b) (i) Which one of these molecules would be found in structure Z?

(1)

- ☒ **A** DNA
- ☒ **B** mRNA
- ☒ **C** tRNA
- ☒ **D** ribose sugar

(ii) Calculate the length of structure Z in micrometres.

[1 micrometre = 1×10^{-6} m]

(3)

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length of structure Z = ... micrometres

(c) Describe what is produced when a cell undergoes mitosis.

(2)

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(Total for Question 7 = 11 marks)

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4. Question 7 · part a.ii · 3 marks · Working and answer

Working and answer

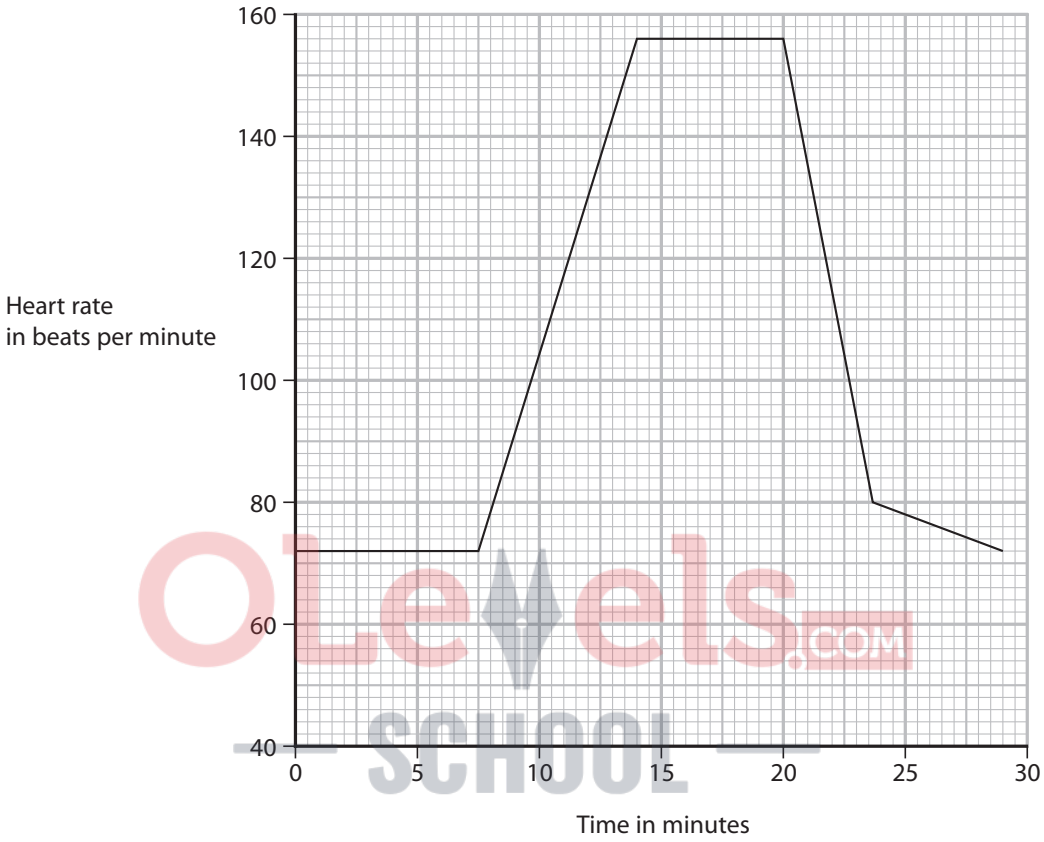
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5. Question 2 · part b.ii · 1 mark

June 2022 Paper 1

Further practice · Attempt part b.ii only.

2 (a) The graph shows how a person's heart rate changes with exercise.



(i) Determine the difference between the maximum heart rate and the resting heart rate.

(2)

difference = ... beats per minute

(ii) Which of these is the time taken for this person to recover from the exercise?

(1)

- ☒ **A** 4 minutes
- ☒ **B** 6 minutes
- ☒ **C** 9 minutes
- ☒ **D** 15 minutes

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5. Question 2 · part b.ii · 1 mark

June 2022 Paper 1

Further practice · Attempt part b.ii only.

(iii) Complete the passage about what happens in the body during exercise using the correct words.

(4)

During exercise ..... gas moves from the lungs into the blood by a process called ..... . This gas is needed by cells for ..... respiration which releases energy. Also during exercise, the amount of urine produced ..... as water is lost in sweat.

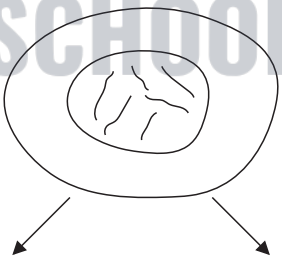
(b) (i) During exercise, body cells get damaged.

Body cells are replaced by a process called mitosis.

The diagram represents one body cell but shows only 6 of the 46 chromosomes.

Complete the diagram to show the new body cells produced from this cell by mitosis.

(2)



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5. Question 2 · part b.ii · 1 mark

June 2022 Paper 1

Further practice · Attempt part b.ii only.

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(ii) Mitosis takes place in four stages.

Which of these gives the stage of mitosis where chromosomes align at the equator of the cell?

(1)

- ☒ **A** anaphase
- ☒ **B** metaphase
- ☒ **C** prophase
- ☒ **D** telophase

(Total for Question 2 = 10 marks)

Working and answer

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6. Question 2 · part b.i · 1 mark

June 2025 Paper 2R

Further practice · Attempt part b.i only.

2 (a) Cells are made up of different components. Each component has a particular function.

Identify the cell component that transports substances inside the cell.

(1)

- ☒ **A** cell membrane
- ☒ **B** cytoplasm
- ☒ **C** endoplasmic reticulum
- ☒ **D** ribosomes

(b) Mitosis is a type of cell division.

A student uses a light microscope to view a cell undergoing four stages in mitosis.

The diagrams, W, X, Y and Z, show the student's observations.



W X Y Z

(Source: https://www.pinterest.co.uk/pin/812547957739062607/)

(i) Give the letters to show the correct order of the diagrams in the process of mitosis.

(1)

.....

(ii) State one function of mitosis in body cells.

(1)

.....

.....

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6. Question 2 · part b.i · 1 mark · Working and answer

Working and answer

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